

Mrinaal Ramachandran

MS in Computer Science - UMass Amherst

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Profile

Graduate student in Computer Science aiming to combine my technical experience and strong interpersonal instincts to build systems that protect the vulnerable and hold abusers accountable. Passionate about applied ML and security, privacy, and applying my skills to relevant and ethical problems!

Education

University of Massachusetts Amherst — M.S. in Computer Science (Expected 2027)

- **Security Concentration** : Adv Computer Network Security, Internet Law, and Digital Forensics.

Arizona State University — B.S. in Computer Science (2022–2025)

- Relevant Coursework: Operating Systems, Distributed Software Development, Computer Architecture, Information Assurance, Computer Network Security, Data Structures and Algorithms, Artificial Intelligence, Natural Language Processing

Experience

Android Security Intern – DriverAI (Startup)

Remote | June 2025 – Aug 2025

- Developed secure authentication workflows using SQLCipher, Play Integrity API, and Credential Manager for Android MVP app.
- Designed AES encryption layer for Room database and implemented encrypted data handling to secure sensitive user input.
- Implemented backend authentication flows including JWT-based token verification and integration with managed identity providers Firebase Auth and AWS Cognito.
- Conducted root detection and SELinux status checks across multiple device profiles, including emulator bypass techniques.
- Collaborated directly with startup CTO to define milestones, propose deliverables, and outline future security tasks.

Undergraduate Research Assistant – PERSUE Labs, ASU

Tempe, AZ | Feb 2024 – May 2025

- Collaborated alongside Professor Hasan to investigate the intricacies of Zoom mini-apps, focusing on data access and security implications.
- Co-authored a research paper submitted to **PETS 2025 (Privacy Enhancing Technologies Symposium)** titled - *Who's Watching You Zoom? Investigating Privacy of Third-Party Zoom Apps* - conducted as part of a year-long longitudinal study analyzing 97,000+ snapshots of Zoom Marketplace apps.
- Developed a proprietary Zoom app, gaining hands-on experience with the Zoom SDK development process and data access / manipulation capabilities
- Conducted privacy and security analysis of Zoom Marketplace apps by reverse-engineering API interactions, auditing OAuth scopes, and Wireshark analysis
- Actively participated in weekly research meetings and paper discussions, developing both technical insights and professional communication skills in a collaborative lab environment.

Projects

Ultra-Wideband (UWB) Data Communication Capstone Project

CSE 485/486 - Computer Science Capstone Project, **General Dynamics Mission Systems (GDMS)**

- Worked alongside fellow seniors and GDMS engineers to complete a two semester capstone project
- Designed and implemented Ultra-Wideband (UWB) communication boards for data transmission, focusing on modular and scalable code to support bidirectional communication between devices
- Conducted in-depth feasibility and hardware testing to validate UWB technology for data transmission, including signal strength, latency, and reliability.
- Collaborated directly with General Dynamics Mission Systems (GDMS) and utilized Agile methodologies to manage sprints, adapt to evolving project scope, and deliver iterative progress

Sarcasm Detection Web Application

- Developed a sophisticated sarcasm detection system utilizing PyTorch and the Hugging Face transformers library, specifically leveraging the distilroberta-base model fine-tuned on over 100,000 data points, achieving an exceptional F1 score of 0.96 on a dataset of 1,500 sarcastic comments derived from movies, TV shows, and daily usage.
- Maintained a robust F1 score of 0.81 on a diverse dataset of 50,000 sarcastic comments sourced from Reddit and Twitter, and achieved an F1 score of 0.84 on sarcastic news headlines—data not seen during training—demonstrating the model's strong accuracy and generalizability across various contexts.
- Engineered a responsive web application with Flask, HTML, jQuery, and AJAX, delivering real-time sarcasm detection with a seamless user experience.
- Applied advanced data preprocessing techniques with pandas to ensure high-quality training data and optimal model performance.
- Overcame challenges such as computational constraints by fine-tuning the model on my local machine, implementing efficient training and evaluation practices to maximize the use of available resources.
- Hosted on AWS EC2 Instance: <http://18.190.155.105>

Publications

Goenka, S., Prabhu, A., Sakurai, P., **Ramachandran, M.**, & Hasan, R. (2025). *Who's Watching You Zoom? Investigating Privacy of Third-Party Zoom Apps*. Submitted to **PETS 2025 (Privacy Enhancing Technologies Symposium)**. Preprint available on arxiv.org/abs/2504.04388

Past Extracurricular Leadership and Achievements

2017 - 2022 | FIRST Tech Challenge - Captain of HS Robotics Team

- Lead programmer & Team Captain of a highly competitive robotics team.
- Developed software using FTC SDK on the Android platform using Java.
- Applied creative problem-solving using engineering principles.
- Implemented computer vision using OpenCV.
- Led the team to and participated in **four state-level** competitions and **two world-level** competitions.
- Managed the division of labor and development plans for the team.

2017 - 2019 | FIRST (For Inspiration and Recognition of Science and Technology) - *Volunteer*

- Over 100 hours per year in volunteering.
- Organized local and regional level competitions by recruiting volunteers.
- Master of Ceremonies for several events.

Additional Information

- Fluent in Java and C, versed in Python, C/C++ and efficient with collaboration tools
- Avid MMA (Mixed Martial Arts) fan, training no-gi jiu jitsu, wrestling, Muay Thai, and Taekwondo
- US Citizen (No work authorization is required)